

Figure 1 is a line graph showing the population fraction of various species as a function of pH. The x-axis is labeled "pH" and ranges from 0 to 14. The y-axis is labeled "population fraction" and ranges from 0.0 to 1.0. The graph displays several curves representing different species:

- A solid purple line starts at a population fraction of approximately 0.88 at pH 0, rises to a plateau of 1.0 between pH 2 and 4, and then decreases to 0.0 by pH 8.
- A dashed red line starts at approximately 0.81 at pH 0, rises to a plateau of approximately 0.92 between pH 2 and 4, and then decreases to 0.0 by pH 8.
- A solid blue line starts at 0.0 at pH 0, rises to a plateau of 1.0 between pH 8 and 11, and then decreases to 0.0 by pH 14.
- A dashed green line starts at 0.0 at pH 0, rises to a plateau of 1.0 between pH 8 and 11, and then decreases to 0.0 by pH 14.
- A dashed orange line starts at approximately 0.08 at pH 0, rises to a plateau of approximately 0.08 between pH 2 and 4, and then decreases to 0.0 by pH 8.
- A solid magenta line remains at 0.0 for the entire pH range from 0 to 14.

